

HealthPulse AI

Full Stack Development Project
Submitted by: Nandana MC

Introduction:

Cardiovascular diseases and metabolic disorders are among the leading causes of death worldwide. Early detection of myocardial infarction (MI) risk and metabolism-related disorders can help prevent severe health complications.

This project proposes a web-based intelligent healthcare platform that:

- Predicts MI risk
- Calculates BMI and metabolism rate
- Monitors health conditions
- Provides health recommendations
- Uses AI and machine learning for prediction

The system allows users to input health parameters and receive instant diagnostic analysis.

Problem Statement:

Traditional health assessment methods:

- Require hospital visits
- Are time-consuming
- Depend heavily on manual diagnosis
- Lack instant accessibility
- May delay early detection

There is a need for:

- Fast
- Automated
- User-friendly
- Real-time health prediction systems

for early identification of heart disease and metabolic abnormalities.

Objectives:

Main Objectives

- Develop an AI-based MI prediction system
- Detect metabolism-related health conditions
- Provide real-time health analysis
- Develop an interactive healthcare website
- Improve early diagnosis and awareness

Specific Objectives

- Calculate BMI
- Predict heart attack risk
- Analyze metabolism rate
- Store user health records
- Generate health reports

Proposed System:

The proposed system is a web-based healthcare platform that integrates:

- Machine Learning
- Health Analytics
- User Authentication
- Real-time Prediction

System Workflow

User Input → Data Processing → AI Prediction → Result Analysis → Health Recommendation

Technology Used:

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python Flask
Database	MySQL
Machine Learning	TensorFlow / Scikit-learn
Visualization	Matplotlib
Deployment	Cloud / Localhost

Literature Survey:

The Implementation of a Body Mass Index (BMI) Calculator in an Android-Based Ideal Body Check and Nutrition Consultation Application”:

summary

This research developed an Android application that calculates BMI and provides nutrition consultation features. The application includes health monitoring, reminders, and progress tracking for users.

Methodology

- User-centered design
- BMI calculation algorithms
- Real-time consultation integration
- Mobile application framework

Advantages

- Easy health monitoring
- Personalized nutrition guidance
- User-friendly interface

“Fitness Tracker (BMI Calculator) Web Application”

Authors: S. R. Kawale et al. (2025)

Summary

The paper presents a web-based BMI tracker using Flask, HTML, CSS, JavaScript, and MySQL. It stores historical BMI records and visualizes trends using charts.

Methodology

- Flask backend
- SQLite/MySQL database
- BMI calculation module
- Interactive data visualization using Chart.js

Advantages

- Historical BMI tracking
- Secure user login
- Interactive dashboard

Mobile Health Applications in Weight Management: A Systematic L



Authors: Katerina Dounavi, Olga Tsoumani (2019)

Summary

This systematic review analyzed mobile health applications used for weight management and obesity control.

Methodology

- Review of 39 research studies
- Comparative analysis of weight management applications
- Behavioral and treatment analysis

Advantages

- Supports healthy food habits
- Encourages self-monitoring
- Improves user engagement

“The Role of Mobile Apps in Obesity Management”

Authors: C. Pujia et al. (2025)

Summary

This study evaluated the effectiveness of smartphone applications in reducing obesity and BMI levels.

Methodology

- Meta-analysis of obesity-management apps
- Statistical comparison of BMI reduction
- Longitudinal observation studies

Advantages

- Significant BMI reduction
- Helps maintain diet discipline
- Improves physical activity tracking

“Developing and Comparing a New BMI Inclusive Energy Expend



Summary

This research proposed a machine learning model to estimate metabolic equivalent tasks (MET) using smartwatch sensor data.

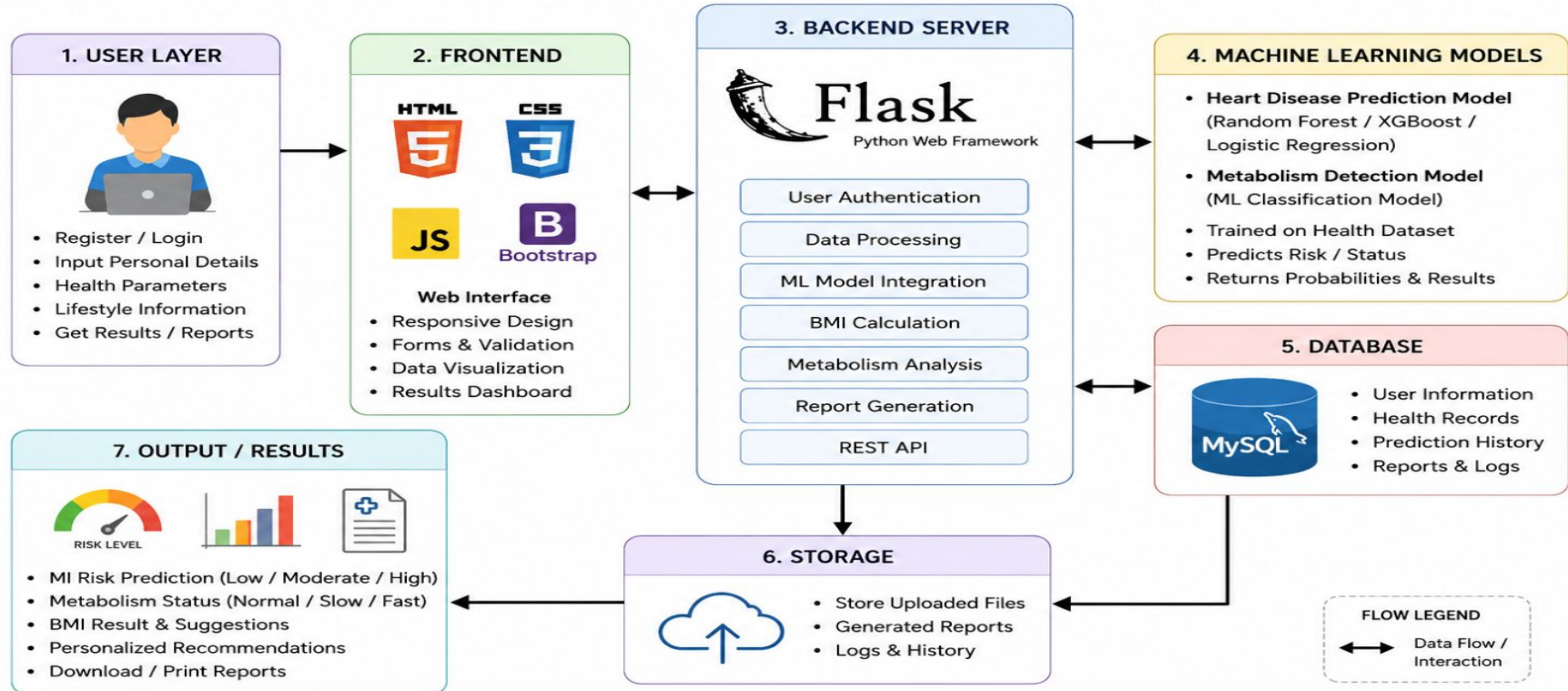
Methodology

- Smartwatch accelerometer data collection
- Machine learning-based metabolism estimation
- Sensor-based activity monitoring

Advantages

- Accurate metabolism prediction
- Real-time monitoring
- Wearable device compatibility

System Architecture:



Proposed System:

The proposed BMI Detection and Metabolism Tracker Website aims to overcome existing limitations by integrating:

- BMI calculation
- Metabolism rate prediction
- AI-based health analysis
- User authentication
- Historical data tracking
- Diet and exercise recommendations
- Interactive dashboard
- Cloud database integration
- Wearable compatibility
- Secure health data management

Conclusion:

The literature survey indicates that BMI and metabolism tracking systems are becoming increasingly important in preventive healthcare. Existing systems provide BMI monitoring and weight management support but lack intelligent personalization and integrated metabolism analysis. The proposed system addresses these gaps using AI-driven analytics, secure web technologies, and real-time tracking features to deliver a comprehensive healthcare solution.

References:

1. Diky Andriansyah, Donny Avianto, “The Implementation of a BMI Calculator in an Android-Based Application,” 2024.
2. S. R. Kawale et al., “Fitness Tracker (BMI Calculator) Web Application,” 2025.
3. Katerina Dounavi, Olga Tsoumani, “Mobile Health Applications in Weight Management,” 2019.
4. C. Pujia et al., “The Role of Mobile Apps in Obesity Management,” 2025.
5. B. Wei et al., “BMI Inclusive Energy Expenditure Model,” 2025.
6. Hera Siddiqui et al., “AI-based BMI Inference from Facial Images,” 2020.